

Supplementary Materials

Crowdstorming You Can Trust? Assessing the Reliability of Crowdsourced Ideation and Prioritization

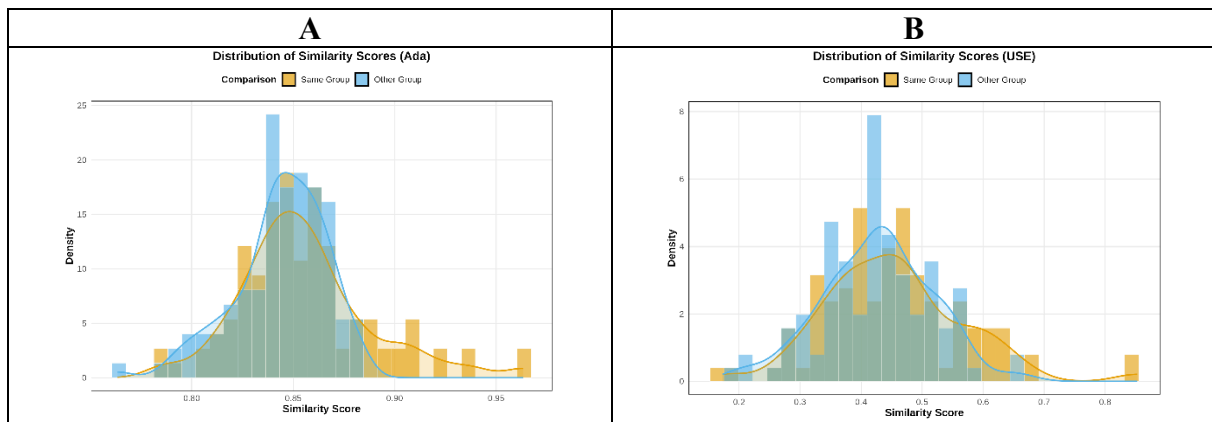
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I. Distribution and diagnostic plots for the ANOVA

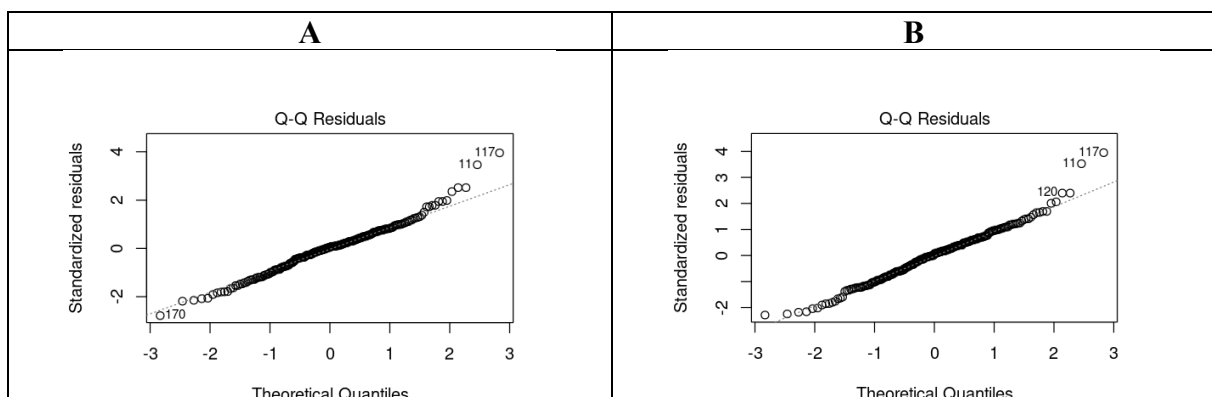
Figure S1. Distribution of Similarity Scores



A. Ada model Distribution of semantic similarity scores using OpenAI's Ada embedding model comparing ideas within the same group (yellow) versus between groups (blue). Cross-group similarities show higher concentration around 0.85, while same-group similarities display greater variance with some very high values (0.90-0.95).

B. USE model Distribution of semantic similarity scores using Google's Universal Sentence Encoder. Despite different absolute values (centered around 0.40-0.45), the pattern is consistent with the Ada model: cross-group similarities show a more concentrated distribution than within-group similarities.

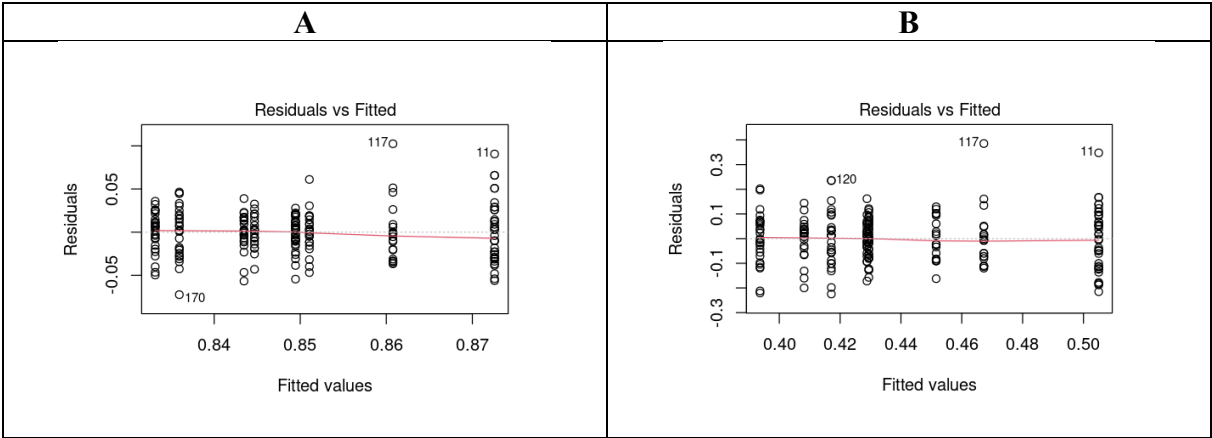
Figure S2. Q-Q Residuals Plots



A. Ada model Q-Q plot assessing normality assumption for ANOVA with Ada similarity scores. Most points follow the reference line with minor deviations at the extremes (observations 170 and 117), indicating the normality assumption is reasonably satisfied.

B. USE model Q-Q plot for ANOVA with USE similarity scores. Similar to the Ada model, most points follow the theoretical normal distribution line with some outliers (observations 117, 11, and 120), supporting adequate normality for analysis.

Figure S3. Residuals vs Fitted Values Plots



A. Ada model Diagnostic plot examining homoscedasticity for the ANOVA model using Ada similarity scores. Residuals show random scatter around zero (-0.05 to 0.05) with no patterns indicating heteroscedasticity, supporting the validity of variance assumptions.

B. USE model Assessment of homoscedasticity for the ANOVA model using USE similarity scores. Residuals display slightly wider range (-0.3 to 0.3) but maintain random scatter across fitted values (0.40-0.50), confirming homoscedasticity assumption.

II. Examples of students' ideas

Table S1. Ideas used to illustrate semantic similarity in Figure 5. Similarity scores are cosine similarities computed using OpenAI's Ada-002 embedding model.

Idea	Group	Full text	Similarity to Idea 39
39 (anchor)	A	Improve Elective Selection and Choice Process: A better point based system to choose electives so that students don't have to choose themselves and check whether their choices have scheduling conflicts.	1.000
98	B	Improve system to select and allocate electives: Improve system to select and allocate electives, giving MBA candidates a real-time opportunity to view his/her scheduling conflicts and number of seats available.	0.924
102	B	Review the list of electives and add more interesting ones: The list of electives needs to be reviewed, new interesting courses could be added.	0.876
95	B	Double bed for rooms for people with partner: Single bed for every type of room, even for people married, the bed is bunk bed.	0.732

III. Sample idea pairs and similarity scores

To illustrate the semantic similarity measurement process, **Table S2** shows two representative examples of ideas and their most similar counterparts across groups. The similarity scores demonstrate how our embedding-based approach quantifies conceptual overlap between ideas.

Table S2. Examples of cross-group idea similarity using different embedding models. Ada-002 refers to OpenAI's text-embedding-ada-002 model, while USE refers to Google's Universal Sentence Encoder.

Idea	Most Similar Counterpart	Ada-002 Similarity	USE Similarity
154. Organize inter-school sporting events: Having an HEC sports tournament involving MBA, MIM and Grade Ecole will increase interaction and hopefully help us perform better at the MBAT	77. More interaction with other programs (EMBA, MIM), other MBA cohorts and other campus (Qatar): HEC Paris is proud of its cultural diversity, if we could interact more with our campus in Qatar virtually or in person during career trek/ by academic or professional activities, it will create more synergy between Europe and the Middle East and more cultural diversity will be realized.	0.858	0.487
6. Individual Study Spaces: Create Individual workspaces for students to study and work throughout S-building. Example is the long corridor on the first floor behind Bellon.	48. Add more rooms or working space: Add more rooms or working space, especially for single person use. For example, there is a limited number of rooms to book and no rooms for single use. If I need to do an online interview, I don't have a place to go.	0.824	0.355

IV. Permutation tests of classification correspondence

To assess whether classification correspondence shown in Figure 7 exceeds what the distribution of classifications alone would produce, we ran permutation tests on the 54 qualifying matched pairs for each embedding model. For each model we computed the proportion of pairs in which an idea and its cross-group counterpart shared a classification, then generated a null distribution by randomly reassigning counterpart classifications across ideas 10,000 times (seed=42). Because this procedure preserves the observed frequency of each class, the resulting null accounts for the uneven distribution of ideas across classifications, including the predominance of High-Priority ideas among qualifying pairs. One-sided p-values report the proportion of permutations in which chance agreement equaled or exceeded the observed value. The same procedure was applied to each class separately.

Table S3. Cross-classification of the 54 qualifying matched pairs. Rows give an idea's classification in its own group; columns give the classification of its most similar counterpart in the other group. Diagonal cells indicate agreement.

A. Ada Model

Own group	HP	Contr	LP	Uncl	Total
<i>High-Priority</i>	17	4	0	2	23
<i>Controversial</i>	6	3	1	3	13
<i>Low-Priority</i>	2	1	0	3	6
<i>Unclassified</i>	3	4	3	2	12
Total	28	12	4	10	54

B. USE Model

Own group	HP	Contr	LP	Uncl	Total
<i>High-Priority</i>	20	3	1	1	25
<i>Controversial</i>	4	2	2	4	12
<i>Low-Priority</i>	3	3	0	2	8
<i>Unclassified</i>	3	1	1	4	9
Total	30	9	4	11	54

Table S4. Permutation test results. Observed agreement is the proportion of pairs sharing a classification; expected agreement is the mean of the permutation distribution.

Model	Class	n	Agree	Observed	Expected	p
Ada	Overall	54	22	40.7%	32.4%	0.102
Ada	<i>High-Priority</i>	23	17	73.9%	51.9%	0.006
Ada	<i>Controversial</i>	13	3	23.1%	22.2%	0.599
Ada	<i>Low-Priority</i>	6	0	0.0%	7.4%	1.000
Ada	<i>Unclassified</i>	12	2	16.7%	18.5%	0.713
USE	Overall	54	26	48.1%	33.9%	0.011
USE	<i>High-Priority</i>	25	20	80.0%	55.6%	0.001
USE	<i>Controversial</i>	12	2	16.7%	16.7%	0.650
USE	<i>Low-Priority</i>	8	0	0.0%	7.4%	1.000
USE	<i>Unclassified</i>	9	4	44.4%	20.4%	0.070

The matched pairs are not fully independent, since two ideas that are each other's closest counterpart contribute two rows to the analysis; the p-values are therefore mildly anticonservative.

The R script implementing these tests, together with the matched-pair data for both models, is available with study data on Github.